

Boundary-Aware Routing for Wind-Power Operating Transitions with an ERA5 Observability Contrast

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Abstract

Wind-power forecast errors become most operationally costly near control transitions, where reserve schedules, ramp-risk alerts, and shortage exposure depend on whether a turbine is still in maximum power point tracking (MPPT) or has entered pitch control. This paper asks a narrow operational question: can a mixture-of-experts (MoE) gate be constrained to identify that operating boundary, and what forecasting and reserve-decision costs does this accountability impose? We build a directed-diffusion gated recurrent unit (GRU) encoder with a node-level MoE gate and constrain the gate with load balancing, regime-anchor alignment, graph smoothness, wake supervision, and boundary-focused forcing. The KDD Cup 2022 wind-farm supervisory control and data acquisition (SCADA) benchmark (WTB) is the main operating-boundary test; ERA5 reanalysis is used as an observability contrast where the thermodynamic marker is already visible through sensible heat flux. The synchronized strict-cache results separate three claims. First, Graph WaveNet and engineering lag-feature baselines provide the accuracy reference: Graph WaveNet reaches overall root mean squared error (RMSE) 225.74 $\pm$ 2.60 and switch-window RMSE 228.12 $\pm$ 3.23 across five seeds, while lag-feature LightGBM and XGBoost reach 227.12 / 232.24 and 227.88 / 233.07 in the same overall/switch metrics. Second, the boundary-forced router pays an RMSE price, with RMSE 236.13 $\pm$ 8.41 and switch-window RMSE 239.86 $\pm$ 8.85, but it recovers the WTB operating partition with gate-regime normalized mutual information/adjusted Rand index (NMI/ARI) 0.8716 $\pm$ 0.0418 / 0.9166 $\pm$ 0.0371. Reloaded checkpoints reproduce the saved metrics; boundary-anchor removal collapses NMI/ARI by 0.7017 / 0.8593, wake-anchor removal is near-null, and temporal, spatial, and global placebos all reduce agreement. Third, reserve value is conditional on the operating window. At shortage-to-reserve cost ratio 10, Graph WaveNet/global-quantile reserve is the main full-sample system baseline, while the boundary-forced gate-bin policy is evaluated as a transition-window diagnostic. On MPPT-to-pitch boundary anchors, that gate-bin policy lowers total cost from 88.13M to 84.58M, violation rate from 0.1038 to 0.0900, and boundary shortage energy from 3.73M to 3.19M relative to the same model's global reserve policy, at the price of 1.85M additional reserve energy. Operator-facing slices show both value and risk: the MPPT-to-pitch slice lowers violation and shortage with more reserve, whereas high-ramp windows expose shortage increases when the gate-conditioned policy carries too little reserve. Gate lead-lag analysis also supports state detection rather than reliable advance warning: agreement with the new regime is 0.3636 three steps before the transition and 0.8008 at the transition. Kelmarsh/Penmanshiel external-site runs fail the pre-specified portability criterion and are reported as boundary-condition diagnostics. Local rated-wind/pitch recalibration recovers only limited test agreement, and a small calibration-window boundary/gate-map adaptation completes but does not restore robust held-out routing. The contribution is physical accountability of routing near wind-turbine operating transitions, with explicit evidence for its forecasting, reserve, and external-site limits.

Keywords: Wind-power operations; operating boundary; reserve decision; mixture of experts; regime routing; spatio-temporal graph learning.

1 Introduction

Average error hides the part of forecasting that operators often care about most. In renewable-rich systems, the costly mistakes cluster in short transition windows, when the local response law changes faster than dispatch, reserves, curtailment, or turbine control can be retuned. In wind farms, this difficulty appears near the maximum power point tracking (MPPT)-to-pitch boundary and during wake-affected inflow shifts, where power stops following wind speed through one smooth mapping [Khodayar and Wang, 2019, Zhou et al., 2024, Haq et al., 2025]. A parallel problem appears in the lower atmosphere: stable-to-convective transitions reorganize transport and surface forcing over short horizons that matter for subsequent prediction [Hersbach et al., 2020]. This paper treats those transition windows as the central forecasting problem.

Spatio-temporal graph forecasting has improved average predictive accuracy in many correlated systems. DCRNN, STGCN, Graph WaveNet, ASTGCN, AGCRN, and MTGNN show that graph propagation and temporal encoding capture asymmetric dependence and dynamic correlation better than dense sequence baselines [Li et al., 2018, Yu et al., 2018, Wu et al., 2019, 2020, Guo et al., 2019, Bai et al., 2020]. Wind-forecasting work has brought the same idea to wake interaction, transport, and turbine coupling [Khodayar and Wang, 2019, Park and Park, 2019, Yu et al., 2020, Daenens et al., 2025]. The evaluation habit has lagged behind the physics. Most benchmarks still reward overall mean absolute error (MAE), root mean squared error (RMSE), or mean absolute percentage error (MAPE), even when the operational failure is concentrated where the mechanism changes [Li et al., 2018, Wu et al., 2019, Khodayar and Wang, 2019, Daenens et al., 2025]. Strong graph encoders improve the map; they do not decide which local map should own a transition.

Mixture-of-experts offers a different answer to heterogeneity. Classical and modern MoE models use routing to allocate different inputs to different predictors, which can increase effective capacity while keeping computation sparse [Jacobs et al., 1991, Jordan and Jacobs, 1994, Shazeer et al., 2017, Fedus et al., 2022]. Recent sequence-forecasting work has started to import the same specialization idea into large time-series models [Shi et al., 2025]. In parallel, physics-guided learning has shown that prior knowledge can enter the pipeline through synthetic pretraining or feature construction. It can also enter through architecture design or physics-based losses and constraints [Karpadne et al., 2017, Read et al., 2019, Park and Park, 2019, Raissi et al., 2019, Karniadakis et al., 2021, Parsa, 2025]. Together, these lines of work show that specialization and physical prior can each help under heterogeneity, but they rarely meet at the routing layer itself.

The gap is at the gate. Regime-transition error is rarely evaluated as a first-class target, although average metrics can hide failure near operating boundaries [Khodayar and Wang, 2019, Wu et al., 2020, Daenens et al., 2025]. Physics-guided models usually regularize latent states or outputs, while the routing decision remains governed by prediction loss [Raissi et al., 2019, Karniadakis et al., 2021, Zehtabiyane-Rezaie et al., 2023]. In control-confounded SCADA streams, that is enough to create a low-error gate with little mechanical meaning. This failure matters because the gate is the part of an MoE that assigns responsibility. If the gate cannot identify the operating boundary, expert specialization becomes a statistical convenience rather than an engineering explanation.

This study puts physical guidance into the routing decision. The model combines a directed-diffusion GRU encoder with a node-level MoE gate and routing losses for expert starvation, regime alignment, local graph consistency, wake exposure, and MPPT-to-pitch ambiguity. WTB and ERA5 are paired because they expose different levels of regime observability: WTB hides the aerodynamic boundary inside control action, while ERA5 makes the thermodynamic marker visible through sensible heat flux. The contribution is deliberately bounded. It reframes non-stationary graph forecasting as operating-boundary routing, places physical supervision at the router, benchmarks the resulting RMSE price against graph, transformer, persistence, power-curve, gradient-boosted lag, and LTSF-style baselines, tests reserve/ramp-risk relevance only in the transition window where the gate has an operational role, and turns the external Kelmars/Penmanshiel result into a boundary-condition experiment that tests whether local rated-wind/pitch recalibration is needed before a WTB boundary rule is reused.

2 Related Work

2.1 Spatio-temporal graph forecasting

Most spatio-temporal graph forecasting studies reach the same empirical result: once spatial dependence is encoded, average forecast error drops. DCRNN, STGCN, and Graph WaveNet established this pattern by combining graph propagation with recurrent or temporal-convolutional encoders [Li et al., 2018, Yu et al., 2018, Wu et al., 2019]. ASTGCN, AGCRN, and MTGNN then relaxed the fixed-graph assumption and improved average accuracy through adaptive dependence learning [Guo et al., 2019, Bai et al., 2020,

Wu et al., 2020]. Wind-forecasting models use graph construction to represent wake interaction, transport, and turbine coupling [Park and Park, 2019, Yu et al., 2020, Daenens et al., 2025]. This literature gives the accuracy baseline that our model must face. It also leaves a routing question open at mechanism transitions.

2.2 Mixture-of-experts and routing for sequence modeling

The MoE literature gives the natural modeling tool for heterogeneous response laws. Early adaptive and hierarchical MoE models showed how routing partitions the input space into regions served by different local experts [Jacobs et al., 1991, Jordan and Jacobs, 1994]. Sparse-routing systems scaled this principle and exposed the familiar training failures: router collapse, expert starvation, and unstable load allocation [Shazeer et al., 2017, Fedus et al., 2022]. Recent sequence-forecasting models have brought the same specialization principle into time-series forecasting [Shi et al., 2025]. For physical regime shifts, a useful router must be judged against mechanism boundaries as well as validation loss.

2.3 Physics-informed and physics-guided learning in spatio-temporal systems

Physics-guided learning has already shown many entry points for prior knowledge. Theory-guided data science and process-guided deep learning use synthetic data, feature construction, pretraining, and post-hoc constraints [Karpatne et al., 2017, Read et al., 2019]. Physics-informed neural network (PINN)-style methods and later reviews extend the idea to optimization, architecture, and losses [Raissi et al., 2019, Karniadakis et al., 2021, Parsa, 2025]. In renewable and atmospheric applications, physics-induced graph neural networks (GNNs), lidar-assisted wind-field prediction, frequency-domain PINNs, and physics-guided wind-farm power models improve robustness or interpretability by injecting physical information into the model pipeline [Park and Park, 2019, Zhang and Zhao, 2021, Li et al., 2025, Zehtabiyani-Rezaie et al., 2023]. This paper uses the prior at the router, where the responsibility assignment is made.

3 Methodology

3.1 Problem setup and notation

The forecasting task is written to separate two decisions that dense models often merge: predicting the future and deciding which local mapping should be active at the anchor time. Let $G = (V, E)$ denote a spatial graph with $N = |V|$ nodes. For each node $i \in V$ and time step t , we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P , the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

Table 1 lists the notation used in the routing and loss definitions. Routing is node-level: each node at each anchor time receives its own gate distribution $\mathbf{g}_{i,t}$. This is essential because nearby turbines or grid cells can move into different local regimes within the same sequence.

3.2 Shared architecture

3.2.1 Physical graph construction

The graph is part of the physical problem specification. In WTB, it exposes likely wake pathways. In ERA5, it preserves geographically local dependence because the thermodynamic regime marker is already

Table 1. Core notation used in the routing and loss definitions.

Symbol	Meaning	Dimension
$\mathbf{x}_{i,t}$	Node feature vector at node i , time t	$\mathbb{R}^F$
$\mathbf{X}_{t-H+1:t}$	Input history window	$\mathbb{R}^{H \times N \times F}$
$y_{i,t}$	Scalar target	$\mathbb{R}$
$\mathbf{h}_{i,t}$	Encoded node context	$\mathbb{R}^d$
$\mathbf{a}_{i,t}$	Physics anchor vector used by the gate	$\mathbb{R}^q$
$\mathbf{z}_{i,t}$	Gate logits	$\mathbb{R}^E$
$\mathbf{g}_{i,t}$	Gate probabilities after softmax	$\mathbb{R}^E$
$f_e(\cdot)$	Expert- e prediction head	$\mathbb{R}^d \rightarrow \mathbb{R}^P$
$R_{i,t}$	Primary regime anchor label	discrete
$W_{i,t}$	Auxiliary wake flag in WTB	$\{0, 1\}$
$\mathcal{A}_t(i, j)$	Edge weight from node j to node i at time t	scalar
E	Number of experts	scalar
K	Top- K used in the balancing loss	scalar

visible in the state. Exact graph constants are listed in the Appendix tables.

For WTB, each turbine has a fixed coordinate $\mathbf{p}_i = (x_i, y_i)$. A candidate set is first built by retaining the k_{nn} nearest turbines within a maximum radius d_{max} . Let $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ and let $\theta_{i,t}$ denote the local meteorological wind-from direction. The downstream unit vector is

$$\mathbf{u}_{i,t} = \begin{bmatrix} \sin(\theta_{i,t} + \pi) \\ \cos(\theta_{i,t} + \pi) \end{bmatrix}.$$

which gives the streamwise and cross-stream distances

$$d_{ij,t}^{\parallel} = \Delta \mathbf{p}_{ij}^{\top} \mathbf{u}_{i,t}, \quad d_{ij,t}^{\perp} = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|.$$

An edge is activated only when turbine j falls inside a downstream cone,

$$\mathbb{I}_{ij,t}^{\text{cone}} = \mathbf{1} \left[d_{ij,t}^{\parallel} > 0 \wedge \arctan \left(\frac{d_{ij,t}^{\perp}}{\max(d_{ij,t}^{\parallel}, 10^{-6})} \right) \leq \phi \right],$$

and the resulting wake weight is

$$\tilde{\mathcal{A}}_t(i, j) = \exp \left(-\frac{d_{ij,t}^{\parallel}}{\alpha} \right) \exp \left(-\frac{|d_{ij,t}^{\perp}|}{\beta} \right) \mathbb{I}_{ij,t}^{\text{cone}},$$

If wind direction is missing, the implementation falls back to a static proximity weight

$$\mathcal{A}^{\text{static}}(i, j) = \exp \left(-\frac{\|\Delta \mathbf{p}_{ij}\|_2}{d_{\text{max}}} \right).$$

Only the strongest M inbound weights are retained for each target and time step. The same weights define a wake score,

$$s_{i,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i, j),$$

which is later used as an auxiliary routing indicator.

ERA5 uses a simpler graph because the main uncertainty is local transport over the retained patch. Each grid point is treated as a node with geographic coordinate (φ_i, λ_i) , and the great-circle distance is computed by the Haversine formula

$$d_{ij} = 2R \arctan\left(\frac{\sqrt{a_{ij}}}{\sqrt{1 - a_{ij}}}\right),$$

where $R = 6371$ km and

$$a_{ij} = \sin^2\left(\frac{\varphi_j - \varphi_i}{2}\right) + \cos(\varphi_i) \cos(\varphi_j) \sin^2\left(\frac{\lambda_j - \lambda_i}{2}\right).$$

We then retain a symmetric k_{nn} -nearest-neighbor graph and define

$$\mathcal{A}(i, j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

where σ is the median retained neighbor distance on the training graph. The design is intentionally simpler than WTB because the ERA5 regime marker is already visible in the observed state.

3.2.2 Directed-diffusion GRU encoder and node-level gate

The encoder is kept modest so that changes in performance can be traced to routing and regularization. Each input window is concatenated with its feature-missingness mask before projection. Two directed diffusion blocks then aggregate self, inbound, and outbound messages on each graph snapshot. Let $\mathbf{X}_t^{(\ell)}$ denote the node state at layer ℓ and time t . A diffusion block updates each node by

$$\mathbf{X}_t^{(\ell+1)} = \text{LN}\left(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)}\right),$$

where Agg_{in} and Agg_{out} are weighted sums over inbound and outbound neighbors. The spatially updated sequence is then processed by a GRU to produce a node-wise context vector $\mathbf{h}_{i,t}$.

Each expert head f_e maps $\mathbf{h}_{i,t}$ to a P -step forecast. Routing remains dense during the forward pass. Every expert contributes, but its contribution is weighted by the soft gate distribution. The gate receives both the learned context and a small physics anchor,

$$\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]), \quad \mathbf{g}_{i,t} = \text{softmax}\left(\frac{\mathbf{z}_{i,t}}{\tau}\right),$$

and the forecast is

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}).$$

WTB uses the anchor

$$\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}, \text{Pab_mean}, s^{\text{wake}}, \text{Patv}],$$

whereas ERA5 uses

$$\mathbf{a}_{i,t}^{\text{ERA5}} = [\text{sshf}, \text{t2m}, \text{wind_speed}, \Delta \text{sshf}].$$

The difference reflects the observability contrast: WTB needs help to recover a partly hidden operating boundary, whereas ERA5 already exposes the relevant thermodynamic marker in the state.

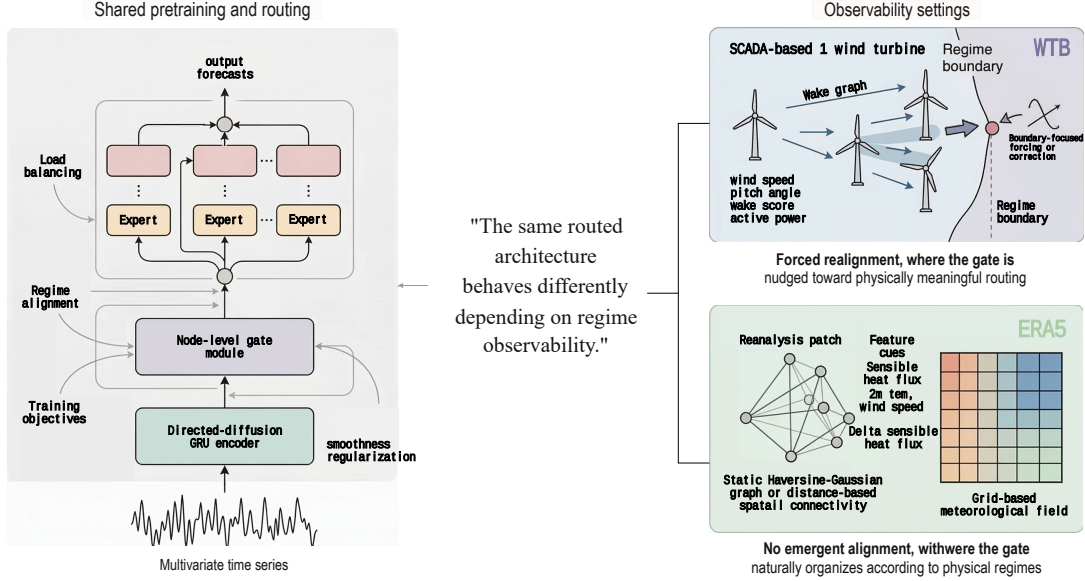


Figure 1: Physics-aligned regime-aware MoE. The two datasets share the same directed-diffusion GRU encoder and node-level MoE routing mechanism. WTB uses a dynamic wake graph and a boundary-focused forcing term, whereas ERA5 uses a Haversine-Gaussian graph and a thermodynamic regime anchor.

3.3 Physics-aligned routing regularizers

Prediction loss does not uniquely identify the gate when several experts can fit averaged dynamics similarly well [Shazeer et al., 2017, Fedus et al., 2022]. WTB makes this failure visible: inflow variation, wake disturbance, and pitch control are observed together, so a low-loss router can still ignore the operating boundary. We use routing regularizers for the specific failure modes that matter here. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}},$$

with inactive terms set to zero in settings where they are not used. Exact weights are listed in the Appendix.

3.3.1 Prediction objective

The prediction loss remains the anchor of the optimization, because a routing correction that stops serving the forecast defeats the purpose of the model. We retain the masked mean-squared-error loss

$$\mathcal{L}_{\text{pred}} = \frac{\sum_{i,t,\tau} m_{i,t+\tau}^y (\hat{y}_{i,t+\tau} - y_{i,t+\tau})^2}{\sum_{i,t,\tau} m_{i,t+\tau}^y},$$

where $m_{i,t+\tau}^y$ is the target-validity mask. All routing regularizers remain subordinate to this prediction objective.

3.3.2 Load balancing

A slightly favored expert can dominate early training and starve the others before regime-specific structure has time to emerge. A balancing term counteracts that early collapse. Let $\mathbf{g}_n$ denote the gate distribution for the n -th valid routed sample in a mini-batch of size B . The soft importance of expert e is

$$I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}, \quad P_e = \frac{I_e}{\sum_{r=1}^E I_r}.$$

The empirical top- K load is

$$f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)].$$

The balancing loss is

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

This term prevents early expert starvation. Regime ownership is supplied by the alignment terms below.

3.3.3 Regime-anchor alignment

Balanced expert usage can still produce a physically meaningless partition. The next term aligns the gate with the coarsest regime structure that can be identified with high confidence. Let $\mathbf{z}_{i,t}^{(1:C)}$ denote the first C gate logits, and let $R_{i,t}$ and $M_{i,t}$ be the primary regime label and its validity mask. The alignment loss is

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}), \quad \Omega = \{(i,t) : M_{i,t} = 1\},$$

with inverse-frequency class weights estimated on the training split. Alignment stabilizes the coarse physical partition only where the labels are reliable.

3.3.4 Boundary-focused forcing (WTB only)

The MPPT-to-pitch boundary in WTB remains ambiguous even after coarse regime alignment because control action partly masks the mechanical transition. A focused forcing term acts only on the most informative MPPT and pitch-control samples. Let

$$Y_{i,t}^{\text{force}} = \begin{cases} 0, & R_{i,t}^{\text{wtb}} = 1, \\ 1, & R_{i,t}^{\text{wtb}} = 2. \end{cases}$$

Using the MPPT-aligned and pitch-aligned expert logits, we define

$$\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE}\left(\begin{bmatrix} z_{i,t}^{(\text{mppt})} \\ z_{i,t}^{(\text{pitch})} \end{bmatrix}, Y_{i,t}^{\text{force}}\right),$$

where $\Omega_{\text{force}} = \{(i,t) : R_{i,t}^{\text{wtb}} \in \{1,2\}\}$. This term concentrates the intervention on the mechanically important boundary.

3.3.5 Wake auxiliary supervision and graph smoothness

After the MPPT-to-pitch boundary is repaired, wake-sensitive samples can still be mixed with non-wake samples inside the same operating regime. A wake auxiliary label identifies that residual ambiguity in WTB, while graph smoothness discourages noisy neighbor-to-neighbor gate jumps in both settings. The wake auxiliary loss is

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\Omega_{\text{wake}}|} \sum_{(i,t) \in \Omega_{\text{wake}}} \text{BCE}(z_{i,t}^{(\text{wake})}, W_{i,t}),$$

where Ω_{wake} contains only MPPT and pitch-control samples for which the wake flag is defined. The smoothness term is

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{i,j} \mathcal{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{i,j} \mathcal{A}_t(i,j)}.$$

The auxiliary term is a WTB-specific repair for wake mixing, while the smoothness term encourages locally coherent routing without prescribing a global partition.

3.4 Dataset-specific anchors and labels

The gate is not supervised everywhere. Instead, it is anchored only where the physical interpretation is clearest, and the ambiguous remainder is organized by prediction loss plus the routing regularizers.

3.4.1 WTB operating regimes and wake labels

In WTB, the primary operating boundary is the transition from MPPT to pitch control. Let

$$\bar{p}_{i,t} = \frac{1}{3} (p_{i,t}^{(1)} + p_{i,t}^{(2)} + p_{i,t}^{(3)}).$$

Using a cut-in threshold u_{idle} , a pitch-transition threshold u_{rated} , and a pitch-angle threshold p_{th} , we define

$$R_{i,t}^{\text{wtb}} = \begin{cases} 0, & \text{if } \text{Wspd}_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & \text{if } u_{\text{idle}} \leq \text{Wspd}_{i,t} \leq u_{\text{rated}} \wedge \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & \text{if } \text{Wspd}_{i,t} > u_{\text{rated}} \wedge \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transition}). \end{cases}$$

Only the first three classes are used in direct alignment. Transition samples are retained for analysis but masked out of label supervision through

$$M_{i,t}^{\text{wtb}} = \mathbf{1}[R_{i,t}^{\text{wtb}} \neq 3].$$

Wake interference is handled separately because it is spatial. Let $q_{0.75}^{\text{wake}}$ denote the upper-quartile threshold of the wake score over valid operating states in the training split. Then

$$W_{i,t} = \begin{cases} 1, & \text{if } s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & \text{if } s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

The exact thresholds are reported in the Appendix tables.

3.4.2 ERA5 thermodynamic regimes

In ERA5, the anchor is built around the stable-to-convective transition. Surface sensible heat flux changes sign across that transition, and large flux gradients mark disturbed periods. Let $\Delta \text{sshf}_{i,t} = \text{sshf}_{i,t} - \text{sshf}_{i,t-1}$. We use a small-margin threshold $\varepsilon_{\text{sshf}}$ for $|\text{sshf}|$. We also use a high-quantile threshold $q_{0.95}^\Delta$ for $|\Delta \text{sshf}|$. The regime label is then defined by

$$R_{i,t}^{\text{era5}} = \begin{cases} 0, & \text{if } \text{sshf}_{i,t} > \varepsilon_{\text{sshf}} \wedge |\Delta \text{sshf}_{i,t}| \leq q_{0.95}^\Delta \quad (\text{convective}), \\ 1, & \text{if } \text{sshf}_{i,t} < -\varepsilon_{\text{sshf}} \wedge |\Delta \text{sshf}_{i,t}| \leq q_{0.95}^\Delta \quad (\text{stable}), \\ 2, & \text{otherwise} \quad (\text{transition/anomalous}). \end{cases}$$

If sensible heat flux is unavailable, the same construction can fall back to an analogous rule based on the temporal gradient of 2 m temperature. The exact thresholds used in the reported experiments are again listed in the Appendix.

4 Experimental Setup

4.1 Datasets and preprocessing

The experiment uses two observability settings, not two interchangeable benchmarks. WTB is the control-confounded case. The KDD Cup 2022 benchmark contains 134 turbines and 245 days of 10 min SCADA measurements [Zhou et al., 2024]. The MPPT-to-pitch transition is only indirectly visible because inflow variation, wake disturbance, and blade-pitch control are mixed in the same stream.

The data are reorganized into synchronous tensors with $T = 35,280$ time steps, $N = 134$ turbines, and $F = 11$ dynamic inputs. Missing inputs are forward-filled within turbine and then completed with training-set means. Invalid or non-positive active power is retained only through its mask so that the model still sees operational irregularity without treating those points as supervised targets.

ERA5 is the signal-expressive case. We use three archived months over a fixed 16×16 hourly patch [Hersbach et al., 2020]. Sensible heat flux and its temporal variation make the stable-to-convective transition directly observable. The retained tensor has $T = 2208$ frames, $N = 256$ nodes, and eight input variables. Both datasets use the same history and horizon lengths, $H = 36$ and $P = 24$, so differences in performance cannot be attributed to unequal context length. The chronological split is 180/30/35 days for WTB and 1325/441/442 frames for ERA5.

4.2 In-family comparison protocol

The comparison has two layers. The first isolates the routing mechanism after shared-capacity explanations have been controlled. The dense baseline uses the same directed-diffusion GRU encoder and replaces the expert mixture with one dense prediction head. The unconstrained MoE adds routed capacity without physical correction. The corrected-routing comparator is dataset-specific: ERA5 uses the full corrected stack, while WTB uses the boundary-focused variant selected for gate recovery. Parameter budgets are matched in WTB and near-matched in ERA5, where the dense model has 103,272 parameters and the routed models have 103,843 parameters.

The second layer tests the RMSE price against stronger and more engineering-facing baselines. Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST are included for WTB, together with deterministic persistence, a fitted physical power-curve baseline, XGBoost and LightGBM lag-feature predictors, and a DLinear-style LTSF baseline. Graph WaveNet, STGCN, PatchTST, TCN, and a deterministic persistence predictor are included for ERA5. This split keeps the paper from using one comparison for two different claims. The in-family layer tests whether physical routing changes the learned partition under a fixed backbone. The baseline layer tests whether the routing intervention remains honest about forecast error when compared with established temporal, graph, and engineering predictors.

Table 2. In-family model comparison used for mechanism validation in the current encoder family.

Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110,012 / ERA5 103,272	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110,012 / ERA5 103,843	Tests whether routed capacity alone is enough
Corrected routing comparator	Node-level soft gate	WTB boundary-forced: $L_{bal} + L_{align} + L_{force}$; ERA5: full corrected stack	WTB 110,012 / ERA5 103,843	Dataset-specific corrected-routing comparator

For compactness, the mechanism tables shorten the three in-family model names to **Dense (matched)**, **Unconstrained MoE**, and **Corrected routing comparator**. The WTB forecasting tables additionally report Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, the full physics-aligned MoE, the WTB boundary-forced router, and engineering baselines built from persistence, power curves, gradient-boosted lag features, and a DLinear-style LTSF readout. The ERA5 forecasting tables additionally report persistence and four learned external baselines. The in-family mechanism families are summarized in Table 2.

4.3 Training protocol and evaluation metrics

All neural models use the same training protocol where the architecture permits it. We use AdamW, early stopping on validation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in the Appendix so that the main text can stay focused on the comparison logic. The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are repeated across five strict-mask seeds. The synchronized strict-cache strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST across seeds 201–205, giving 20 completed baseline runs under the same strict anchor-valid cache. The engineering WTB baselines are deterministic or sampled single-run readouts from the same strict cache and are used to anchor practical forecast difficulty rather than to make a seed-level neural ranking. ERA5 main in-family models are repeated across five seeds, and the ERA5 learned baselines are repeated across three seeds. Thresholds and routing weights are set before test evaluation and then audited by sensitivity sweeps and boundary negative controls, so the main routing result is not selected by test-set NMI.

The metrics follow the claim. Overall MAE and RMSE measure forecasting accuracy. Switch-window MAE and RMSE focus on mechanism changes. Regime-wise RMSE locates the error reduction. Normalized mutual information (NMI), adjusted Rand index (ARI), usage entropy, and confusion matrices test whether the gate corresponds to physically interpretable structure.

Figure 2 gives the physical setting before the forecasting results are introduced.

5 Results

5.1 Overall performance

WTB delivers a hard accuracy result: the strongest graph and engineering forecasters beat the routed family under the synchronized strict-cache refresh. Graph WaveNet reaches overall RMSE 225.74 +/- 2.60 and switch-window RMSE 228.12 +/- 3.23 across seeds 201–205. PatchTST reaches 228.07 +/- 4.05 and 231.22 +/- 4.15. Graph Transformer reaches 235.38 +/- 5.15 and 237.06 +/- 5.78, while GAT-

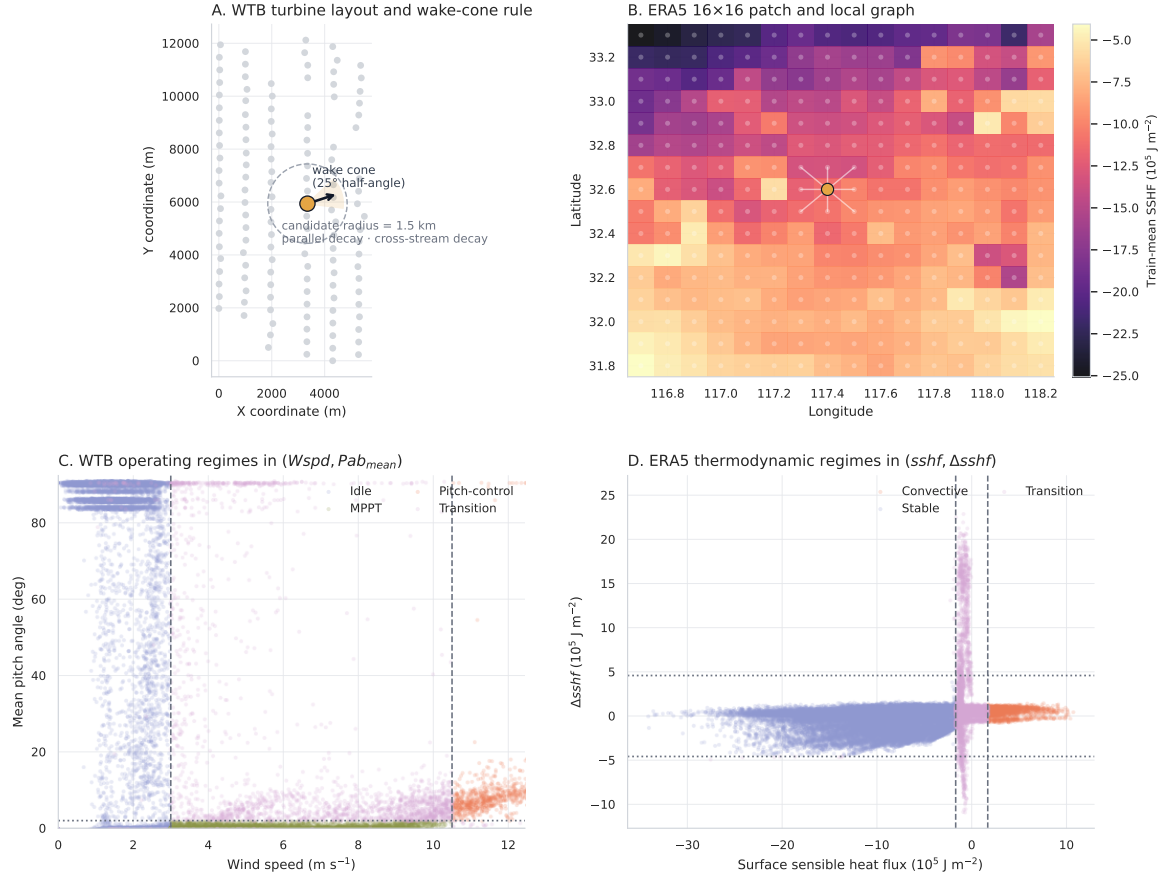


Figure 2: Dataset geometry and physical regime anchors. (A) WTB turbine layout with the schematic wake cone and the retained candidate radius used in the dynamic directed wake graph. (B) ERA5 16x16 patch with training-mean sensible heat flux and local Haversine-Gaussian graph connections around the central node. (C) WTB operating regimes in the ($Wspd, Pab_{mean}$) plane with fixed operating-rule boundaries. (D) ERA5 thermodynamic regimes in the ($sshf, \Delta sshf$) plane with thresholds estimated from the training split.

Model	n	Overall RMSE	Switch RMSE	NMI	ARI
GAT-GRU	5	236.59 $\pm$ 7.80	240.17 $\pm$ 8.50	–	–
Graph Transformer	5	235.38 $\pm$ 5.15	237.06 $\pm$ 5.78	–	–
Graph WaveNet	5	225.74 $\pm$ 2.60	228.12 $\pm$ 3.23	–	–
PatchTST	5	228.07 $\pm$ 4.05	231.22 $\pm$ 4.15	–	–
MoE + L_bal + L_align + L_force	5	236.13 $\pm$ 8.41	239.86 $\pm$ 8.85	0.8716 $\pm$ 0.0418	0.9166 $\pm$ 0.0371

Table 1. Strict-cache WTB leaderboard after synchronized external-baseline refresh.

Table. WTB engineering forecasting baselines evaluated on the strict cache.

Baseline	Overall RMSE	Switch RMSE	Pitch-control RMSE
Persistence	243.23	246.61	262.35
Physical power curve	241.02	245.37	299.30
XGBoost lag-feature	227.88	233.07	277.33
LightGBM lag-feature	227.12	232.24	270.23
DLinear-style LTSF	246.81	251.15	284.06

GRU reaches 236.59 $\pm$ 7.80 and 240.17 $\pm$ 8.50. Among engineering baselines, LightGBM lag features reach overall/switch RMSE 227.12 / 232.24, XGBoost lag features 227.88 / 233.07, a fitted physical power curve 241.02 / 245.37, persistence 243.23 / 246.61, and the DLinear-style LTSF readout 246.81 / 251.15. The boundary-forced router reaches overall RMSE 236.13 $\pm$ 8.41, switch-window RMSE 239.86 $\pm$ 8.85, and pitch-control RMSE 286.95 $\pm$ 22.07 across seeds 201–205. The completed strict-cache leaderboard places the routed model outside a headline-error objective and motivates the narrower routing-recovery claim.

ERA5 makes the same separation sharper. A last-value persistence baseline gives the lowest headline error, with overall RMSE 0.7127 and switch-window RMSE 0.6814. Among learned ERA5 baselines, Graph WaveNet is best at 0.8417 $\pm$ 0.0540, followed by the matched dense encoder at 0.8641 $\pm$ 0.0690, physics-aligned MoE at 0.8735 $\pm$ 0.0479, TCN at 0.8784 $\pm$ 0.0286, PatchTST at 0.8817 $\pm$ 0.0675, Unconstrained MoE at 0.9294 $\pm$ 0.0451, and STGCN at 0.9893 $\pm$ 0.1344. The forecasting tables set the terms of the paper: physics-aligned routing earns its value through interpretable responsibility. That value is only credible if the routing metrics improve under the same evaluation protocol.

The two WTB tables below are the submission-facing forecast evidence: the strict-cache neural leaderboard and the strict-cache engineering baseline panel. Both are tied to the final evidence package and avoid pre-strict-mask corrected-router rows.

5.2 Transition-window and regime-slice performance

The difficult slices repeat the same separation between forecasting accuracy and routing evidence. In the completed strict-cache baseline rows, Graph WaveNet reaches switch-window RMSE 228.12 $\pm$ 3.23 and pitch-control RMSE 282.67 $\pm$ 31.41. PatchTST reaches switch-window RMSE 231.22 $\pm$ 4.15 and pitch-control RMSE 277.21 $\pm$ 19.30. Graph Transformer reaches switch-window RMSE 237.06 $\pm$ 5.78 and pitch-control RMSE 280.50 $\pm$ 35.98, while GAT-GRU reaches switch-window RMSE 240.17 $\pm$ 8.50 and pitch-control RMSE 318.28 $\pm$ 49.88. Under the strict anchor-valid mask, the boundary-forced router reaches pitch-control RMSE 286.95 $\pm$ 22.07 and switch-window RMSE 239.86 $\pm$ 8.85. These slice results support a mechanism-evidence update, not a headline-error claim. ERA5 again favors persistence. Among learned ERA5 models, Graph WaveNet gives the best overall and switch-window RMSE, while physics-aligned routing improves over Unconstrained MoE in the transition slice without beating the dense model or persistence on headline error. This is calibration around a visible thermodynamic boundary, not broad forecast repair.

The submission-facing slice evidence is therefore restricted to strict-cache rows and the boundary-slice

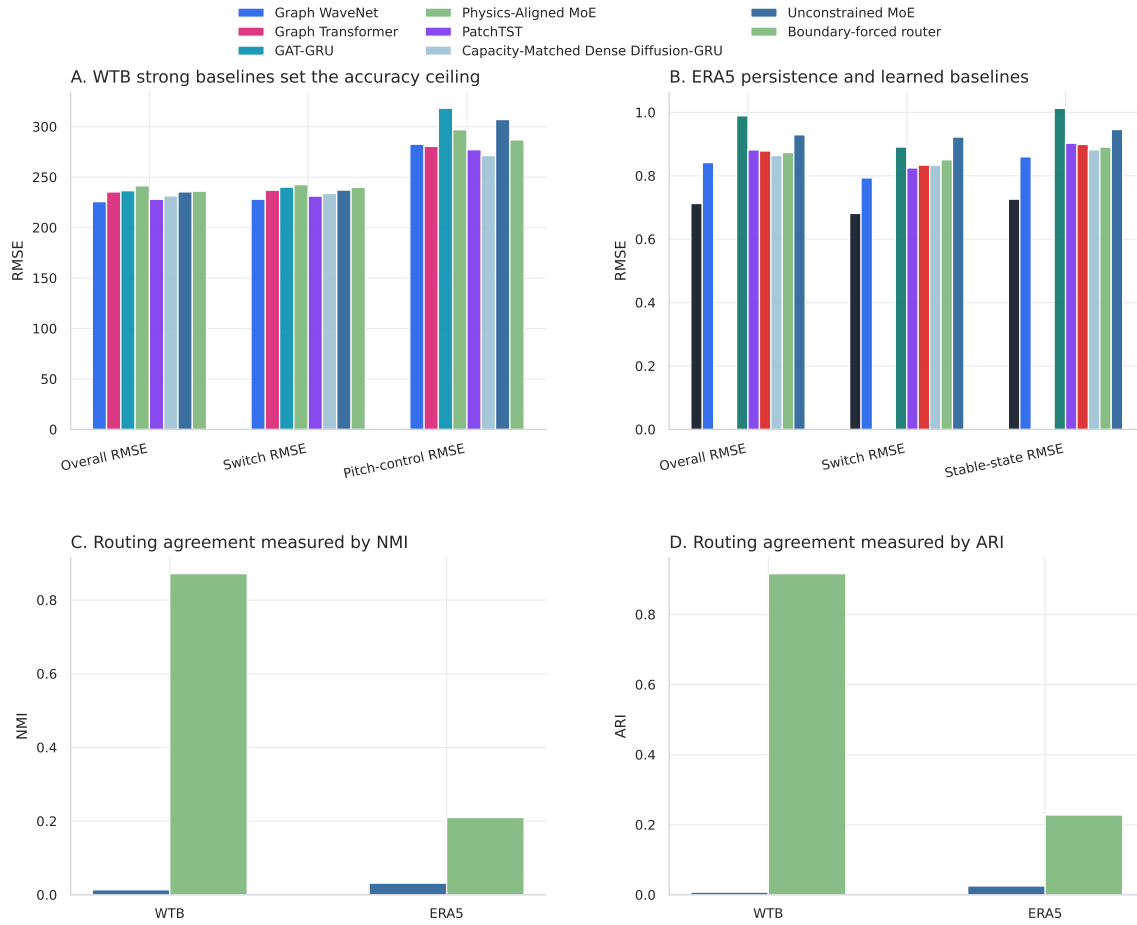


Figure 3: Summary of forecasting and routing outcomes. Panels A and B compress the learned-model error comparison for WTB and ERA5, with ERA5 persistence shown as a deterministic reference. In WTB, the displayed corrected comparator is the boundary-forced routing variant; the deeper full stack is reported separately. Panels C and D summarize routing agreement through NMI and ARI.

Table 2. Strict-mask WTB MPPT-to-pitch boundary-slice audit.

Slice	RMSE	NMI	ARI	Anchor cells
boundary_band	321.1711 +/- 21.3067	0.6562	0.7766	7879
nonboundary_valid	254.0981 +/- 8.6588	0.6326	0.6056	283797
mppt_core	253.7099 +/- 8.7181	NA	NA	277719
pitch_core	271.1653 +/- 16.9980	NA	NA	6175

Table 3. Strict-mask WTB boundary-band difficulty relative to comparator slices.

Comparison	Delta RMSE	Delta MAE
boundary_band_minus_nonboundary_valid	67.0731 [52.5408, 84.2072]	81.3676 [65.5070, 97.2282]
boundary_band_minus_mppt_core	67.4612 [52.6902, 82.9221]	82.3750 [66.2215, 99.9064]
boundary_band_minus_pitch_core	50.0058 [41.5782, 56.4701]	36.7713 [30.7509, 42.8743]

audit below. We do not mix pre-strict corrected-router rows with the current strict-cache evaluation.

We therefore add a stricter boundary-slice audit directly from the saved strict-mask predictions and the strict cache physics. The audit isolates anchors within $\pm 1.0 \text{ m s}^{-1}$ of the rated-wind MPPT-to-pitch boundary and compares them with valid non-boundary MPPT/pitch anchors and with single-regime core slices. The boundary band is the hardest forecasting slice: RMSE is 321.17 +/- 21.31, which is 67.07 higher than the non-boundary valid slice (95% bootstrap interval 52.54 to 84.21), 67.46 higher than the MPPT core, and 50.01 higher than the pitch-control core. At the same time, the boundary band still carries mixed-regime routing structure, with NMI/ARI 0.6562 / 0.7766. This closes a narrower mechanism question: the strict-mask gate is not only globally aligned, but remains measurable at the operating boundary where prediction is hardest.

5.3 Routing evidence

Routing recovery is the positive result. Across five ERA5 seeds, Unconstrained MoE keeps high expert-usage entropy but near-zero routing agreement, with NMI/ARI of 0.0317 +/- 0.0219 / 0.0253 +/- 0.0626. Physics-aligned MoE raises those scores to 0.2100 +/- 0.0972 / 0.2279 +/- 0.1296. This is a modest but useful correction because the ERA5 marker is already visible in the input state. WTB shows the stronger effect after the strict anchor-valid rerun: the boundary-forced router reaches gate-regime NMI/ARI of 0.8716 +/- 0.0418 / 0.9166 +/- 0.0371 across seeds 201–205. The previous weak seed-204 behavior disappears under the strict cache, with seed 204 reaching NMI/ARI 0.8591 / 0.9060. Explicit gate correction therefore recovers a physically meaningful partition in the control-confounded setting, while the result remains bounded to the tested WTB/ERA5 settings.

The current WTB mechanism table above is the submission-facing source for the corrected router. ERA5 routing values are used only as the observability contrast already summarized in the text and final evidence package.

Figure 4 makes the WTB repair visible in the operating plane. After correction, the dominant expert regions align with the MPPT and pitch-control structure, and the confusion matrices contract around the intended operating partition. The corrected gate becomes readable at the boundary that matters operationally, with the RMSE cost already quantified above.

Figure 5 traces the same contrast through time. In the WTB switch window, the boundary-forced router carries responsibility through the MPPT-to-pitch transition more coherently than the unconstrained gate. In ERA5, expert weights evolve with sensible heat flux and regime shading, as expected when the marker is already expressed in the state. The case studies tie the routing metrics to the physical mechanisms

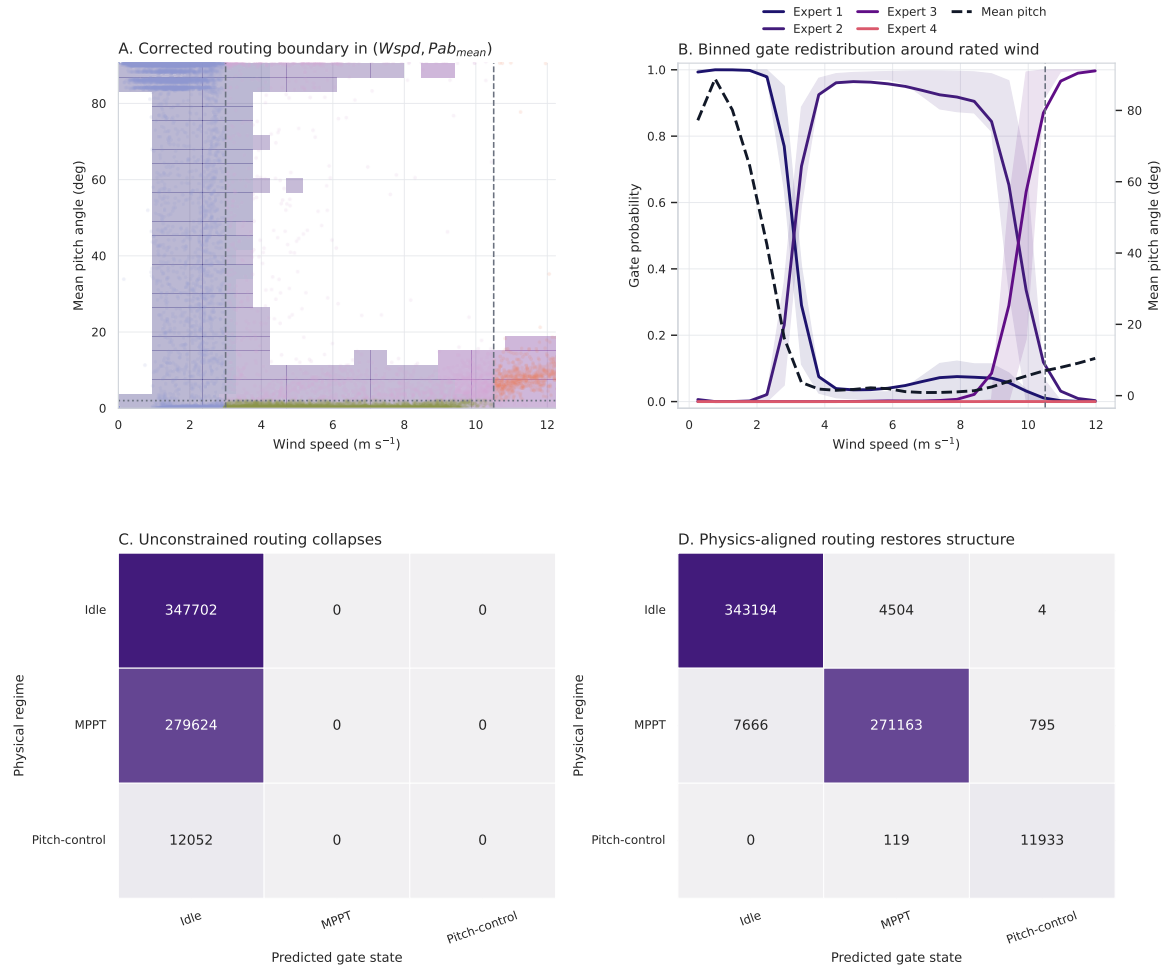


Figure 4: WTB routing diagnostics. Panel A maps the dominant expert in the $(Wspd, Pab_{mean})$ plane for the boundary-forced routing variant, with the sample cloud colored by physical operating regime. Panel B shows the binned gate redistribution against wind speed together with the mean pitch curve. Panels C and D compare the unconstrained and corrected confusion matrices.

Table 4. Strict-anchor-mask WTB five-seed metrics.

Seed	Overall RMSE	Switch RMSE	Pitch RMSE	NMI	ARI
201	245.0565	246.9875	308.0279	0.8685	0.9242
202	244.3905	248.7028	281.8781	0.9377	0.9692
203	228.0898	229.0400	269.9951	0.8221	0.8658
204	227.7822	232.0819	263.0312	0.8591	0.9060
205	235.3335	242.4639	311.8080	0.8708	0.9177
Mean +/- SD	236.1305 +/- 8.4096	239.8552 +/- 8.8508	286.9480 +/- 22.0652	0.8716 +/- 0.0418	0.9166 +/- 0.0371

behind them.

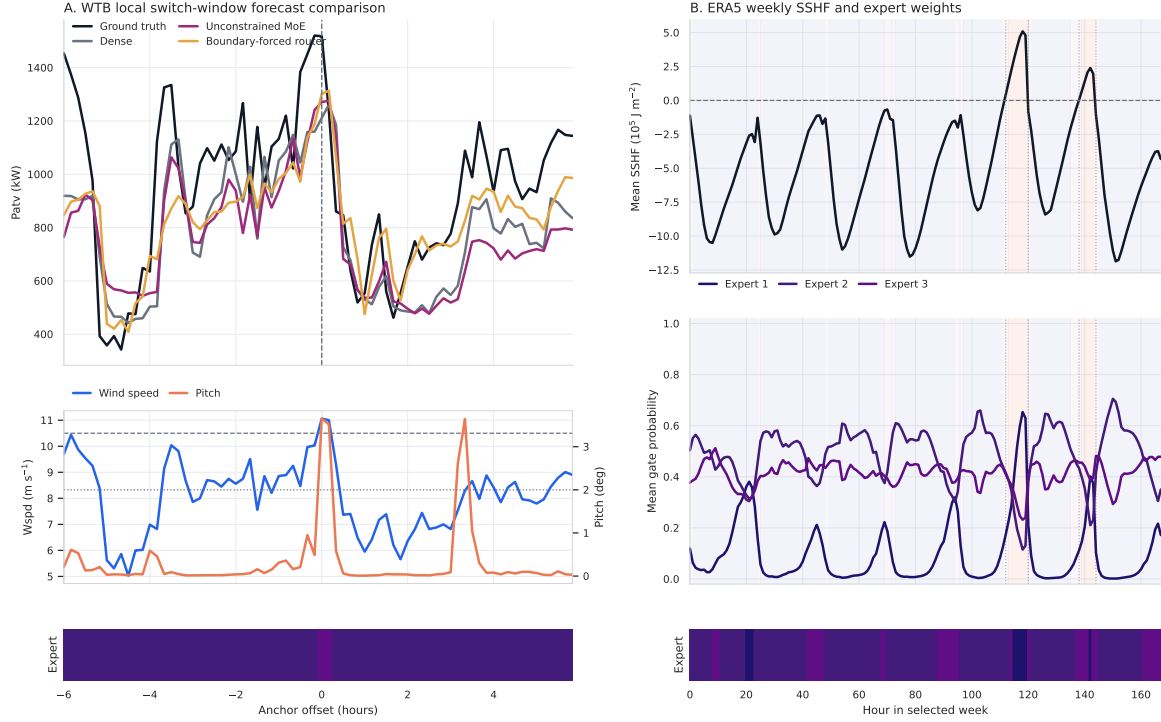


Figure 5: Dual case studies for correction and emergence. Panel A shows a local WTB switch window in which the boundary-forced routing variant remains stable while the dense and unconstrained models deviate more strongly near the MPPT-to-pitch transition; wind speed, pitch angle, and the dominant expert strip are plotted underneath. Panel B shows the ERA5 positive-control view, where sensible heat flux, regime shading, and expert weights evolve coherently over a selected week.

5.4 Operational value and failure cases

The routing claim needs an energy-system consequence, not only a semantic score. We therefore turn the earlier reserve proxy into a conditional operating audit: from the saved WTB validation and test predictions, each model calibrates reserve levels on validation shortfall and is charged reserve cost plus a shortage penalty on test shortfall. The audit reports the quantities a reserve desk would inspect—total cost, violation rate, reserve energy, and shortage energy—and deliberately asks a local question. Does a gate-aware reserve rule change shortage exposure where the gate is supposed to matter, namely MPPT-to-pitch boundary anchors within $\pm 1.0 \text{ m s}^{-1}$ of rated wind? We count this as operating value only when the gate-aware policy improves total cost, violation rate, and shortage energy together in that boundary window.

At the main shortage-to-reserve cost ratio of 10, Graph WaveNet/global-quantile reserve is the principal energy-system baseline, with mean full-sample total cost 4.6407×10^8 ; Graph WaveNet/physical-bin

reserve is reported alongside it as a physics-stratified system comparator. The boundary-forced router is not the cheapest full-sample reserve system. Its operational value appears in the boundary window. On boundary anchors, moving the same boundary-forced model from global reserve to a gate-bin reserve policy lowers mean total cost from 88.13M to 84.58M, violation rate from 0.1038 to 0.0900, and boundary shortage energy from 3.73M to 3.19M, while increasing reserve energy from 50.82M to 52.67M. The physical-bin policy is an oracle-style diagnostic and is slightly lower still, with 83.78M total cost and 3.05M shortage energy. On the full sample and non-boundary subset, however, the gate-bin policy can reduce reserve expenditure and total cost while raising violation and shortage relative to the same model's global policy. The system claim is therefore not that the gate produces a globally better reserve policy; it is that operating-boundary accountability changes the reserve-shortage tradeoff in the window where dispatch risk is physically concentrated.

Cost-ratio sensitivity makes the same boundary explicit. At ratio 2, the penalty is too low to buy reserve and the gate-bin boundary policy is identical to global reserve. At ratios 5 and 10, the boundary-forced gate-bin policy lowers same-model boundary shortage and violation relative to global reserve, with the ratio-10 operating point reducing shortage from 3.73M to 3.19M. At ratio 20 the advantage narrows to a small shortage reduction with almost unchanged violation. At ratio 50 the same-model global policy is safer and cheaper. This is the Applied Energy reading of the reserve result: the gate is a conditional reserve-allocation signal around the MPPT-to-pitch boundary, not a replacement for system-level stochastic reserve optimization.

We also export operator-facing slices that are closer to reserve desk language than the aggregate boundary/non-boundary split. The four slices are MPPT-to-pitch transitions, pitch-to-MPPT transitions, validation-calibrated high absolute ramp, and validation-calibrated low absolute ramp. For the boundary-forced router, gate-bin reserve reduces violation and shortage energy in the MPPT-to-pitch slice relative to the same model's global policy, but it pays an added reserve-cost price: 17.93M total cost, 0.1027 violation rate, and 0.69M shortage energy, compared with 16.33M, 0.1124, and 0.77M under global reserve. The low-ramp slice shows the cleaner cost/risk improvement, with gate-bin total cost 10.73M versus 11.24M and shortage energy 0.40M versus 0.44M. The pitch-to-MPPT and high-ramp slices are not gate-bin wins. In the high-ramp slice, gate-bin reserve reduces total cost slightly but raises violation from 0.0766 to 0.1253 and shortage energy from 1.06M to 2.51M because it carries much less reserve energy. This is why the reserve claim remains a boundary-window decision diagnostic rather than a global reserve-policy claim.

The three operational cases in the next table are included to prevent a common misreading. A correct gate can help allocate attention around the MPPT-to-pitch boundary, but it can also be misleading when the downstream forecast is bad or when a low-reserve gate-bin policy is applied to high-ramp windows.

Lead-lag analysis gives the same operational boundary. Around physical regime switches, the correct-ed gate matches the new regime at rate 0.4066 six steps before the switch and 0.3636 three steps before it, then rises to 0.8008 at the switch and remains 0.6313/0.6094 three/six steps after it. This does not support an early-warning claim. It supports a narrower state-detection claim: the gate is most reliable once the transition is present, so the reserve experiment should be framed as boundary-aware reserve/ramp-risk conditioning rather than as advance prediction of the switch.

The reviewer-statistics pack reports the complementary failure mode. It contains 24 gate-correct high-error cases and 24 boundary gate-correct high-error cases, and the full lists are retained in the supplement. These rows are important because they separate the two parts of the model. In those samples, the router assigns the sample to the physically appropriate operating regime, but the expert forecast is still numerically poor. The usual pattern is a value-map failure near high-error boundary windows, not a collapse of mechanism responsibility. We therefore treat these cases as evidence for the paper's boundary condition: correct routing is necessary for mechanism interpretation, but it is not sufficient for accurate power prediction.

Table. Operator-facing cases: when the boundary gate helps and when it misleads.

Operational case	Evidence	System reading	Claim boundary
MPPT-to-pitch boundary reserve	Boundary window: total cost 88.13M -> 84.58M, violation 0.1038 -> 0.0900, shortage 3.73M -> 3.19M, reserve 50.82M -> 52.67M. MPPT-to-pitch slice: violation 0.1124 -> 0.1027 and shortage 0.77M -> 0.69M, with total cost 16.33M -> 17.93M.	Gate-bin reserve is useful as a boundary shortage alert, especially when the turbine enters pitch control, but the transition slice pays extra reserve cost.	Supported only as a boundary-window diagnostic, not as a full-sample reserve policy.
High-ramp shortage risk	Gate-bin total cost 67.08M versus 67.67M global, but violation rises 0.0766 -> 0.1253, shortage rises 1.06M -> 2.51M, and reserve energy falls 57.03M -> 42.02M.	A gate-conditioned rule can look cheaper by carrying less reserve while exposing the system to more shortage energy in high-ramp windows.	Diagnostic failure case: use ramp-specific checks before deploying gate-bin reserve.
Gate-correct but forecast-bad	Gate label = pitch, true regime = pitch, confidence = 0.99999, Wspd = 11.30, Pab_mean = 85.72, target mean = 79.18 kW, prediction mean = 1336.76 kW, RMSE = 1258.50.	The router can assign the right operating responsibility while the expert value map is wrong; gate correctness is not a forecast-accuracy guarantee.	Supported as failure transparency and accountability, not as dispatch-ready point forecasting.

Table. Boundary-window reserve audit for the WTB boundary-forced router at shortage-to-reserve cost ratio 10.

Reserve policy	Total cost	Violation rate	Reserve energy	Boundary shortage
Global	88.13M	0.1038	50.82M	3.73M
Physical-bin	83.78M	0.0880	53.26M	3.05M
Gate-bin	84.58M	0.0900	52.67M	3.19M

5.5 Spatial and future holdout stress tests

The reviewer-facing spatial holdout tests whether the routing evidence survives when entire turbines are unavailable during training and validation. The east-node protocol holds out 27 of 134 turbines (20.15%). During training and validation, target and regime-valid entries for holdout nodes are masked; during testing, train-node targets are masked and evaluation is restricted to the held-out nodes. The guard reports `mask_policy_pass = true`, zero strict-anchor violations, 15/15 completed runs, and a passed mechanism gate. In this harder within-farm spatial transfer setting, the boundary-forced router reaches overall RMSE 233.87 +/- 2.84, switch-window RMSE 234.87 +/- 2.33, and NMI/ARI 0.8340 +/- 0.0958 / 0.8829 +/- 0.0875. Context-supervised routing is more accurate but less semantically aligned, with NMI/ARI 0.7479 +/- 0.0461 / 0.8084 +/- 0.0717. Anchor-only routing is semantically strong but less stable. The spatial intervention audit keeps the same interpretation as the in-sample strict-mask mechanism test: boundary-anchor removal causes a large NMI/ARI loss and a positive RMSE penalty, while wake-anchor removal remains near-null. This is not an out-of-site wind-farm result; it is a within-farm node-held-out stress test that checks whether the learned routing semantics extrapolate to unseen turbines.

The future-period holdout uses a pre-specified later WTB period with seeds 301–305 and evaluates only the holdout split. All 5/5 runs complete, the mechanism gate passes, and the downstream holdout mechanism intervention, placebo, boundary-slice, and reviewer statistics packs are present. The boundary-forced router reaches holdout overall RMSE 235.49 +/- 7.19, switch-window RMSE 239.93 +/- 7.27, and NMI/ARI 0.8352 +/- 0.0669 / 0.8870 +/- 0.0669. Together with the spatial holdout, this supports a bounded claim: the evidence is within-farm spatial and temporal stress validation of routing semantics, not a forecast guarantee and not evidence that the rule carries to a different wind farm without local testing.

Table. Boundary-window operational reserve slices at shortage-to-reserve cost ratio 10.

Slice	Policy	Cost	Violation	Reserve	Shortage	Valid cells
mppt_to_pitch	Boundary-forced router/global	16.33M	0.1124	8.62M	0.77M	1968
mppt_to_pitch	Boundary-forced router/physical-bin	16.36M	0.0936	10.72M	0.56M	1968
mppt_to_pitch	Graph WaveNet/physical-bin	17.43M	0.0981	10.83M	0.66M	1968
mppt_to_pitch	Graph WaveNet/global	17.55M	0.1109	9.33M	0.82M	1968
mppt_to_pitch	Boundary-forced router/gate-bin	17.93M	0.1027	11.02M	0.69M	1968
pitch_to_mppt	Boundary-forced router/global	20.27M	0.0604	17.38M	0.29M	1992
pitch_to_mppt	Boundary-forced router/physical-bin	20.65M	0.0772	17.16M	0.35M	1992
pitch_to_mppt	Boundary-forced router/gate-bin	20.83M	0.0833	16.72M	0.41M	1992
pitch_to_mppt	Graph WaveNet/global	21.26M	0.0982	15.49M	0.58M	1992
pitch_to_mppt	Graph WaveNet/physical-bin	21.59M	0.1047	15.21M	0.64M	1992
high_ramp	Graph WaveNet/physical-bin	65.83M	0.1224	45.76M	2.01M	6212
high_ramp	Boundary-forced router/gate-bin	67.08M	0.1253	42.02M	2.51M	6212
high_ramp	Graph WaveNet/global	67.15M	0.0965	51.67M	1.55M	6212
high_ramp	Boundary-forced router/global	67.67M	0.0766	57.03M	1.06M	6212
high_ramp	Boundary-forced router/physical-bin	68.02M	0.1065	49.96M	1.81M	6212
low_ramp	Boundary-forced router/gate-bin	10.73M	0.0845	6.70M	0.40M	5594
low_ramp	Graph WaveNet/physical-bin	10.84M	0.0822	6.85M	0.40M	5594
low_ramp	Graph WaveNet/global	10.88M	0.0817	6.95M	0.39M	5594
low_ramp	Boundary-forced router/physical-bin	11.15M	0.0837	6.79M	0.44M	5594
low_ramp	Boundary-forced router/global	11.24M	0.0849	6.84M	0.44M	5594

5.6 Ablation

The WTB ablation exposes a controllable accuracy-semantics frontier. The completed strict-cache 50-run ablation covers dense, unconstrained, balance-only, align-only, balance-align, boundary-forced, s-smoothed boundary-forced, context-aligned, anchor-only, and full variants across seeds 201–205, with the routing-control mechanism gate passed for the three routing variants. The submission-facing reading is deliberately paired: every ablation row must report RMSE together with NMI/ARI. With load balancing, regime alignment, and boundary forcing active, the selected corrected router reaches NMI/ARI 0.8716 +/- 0.0418 / 0.9166 +/- 0.0371. Boundary forcing remains an interpretability intervention with a measured forecasting price rather than a free accuracy improvement.

The ablation claim is therefore the shape of the frontier, not a ranking headline. The current strict-cache guard records 50/50 expected runs complete and 15/15 routing-gate audited runs passed; the paper uses that guard to keep the ablation as evidence for the RMSE/NMI tradeoff.

Figure 6 visualizes this frontier. Unconstrained MoE sits in the low-semantics region. L_{align} moves the gate sharply upward with a modest RMSE change. The force/smooth variants continue toward cleaner partitions while moving rightward on RMSE. The full physics-aligned MoE is the routed forecasting compromise; the boundary-forced variants occupy the semantic-repair end of the frontier.

5.7 Robustness and fairness checks

The robustness and fairness checks rule out two easy explanations for the WTB result. The in-family comparisons use multiple strict-mask seeds, and the routed in-family variants share the same parameter count, making a pure capacity explanation unlikely. The completed strict-cache baseline refresh sharpens the forecasting check: Graph WaveNet is lower than the boundary-forced router by about 10.39 RMSE on average, PatchTST is lower by about 8.06 RMSE, and Graph Transformer and GAT-GRU are close in error but do not change the routing-accountability result. The reviewer pack therefore supports the same qualitative reading as the mechanism tables: the method earns its value through gate-regime responsibility, not headline error dominance. The corresponding ERA5 RMSE difference between physics-aligned

Table. Boundary-window cost-ratio sensitivity in reserve-system language.

Cost ratio	Graph WaveNet reference	Boundary router: gate-bin vs global	Reading
2	Graph WaveNet/global: cost 27.82M, violation 0.3237, shortage 13.91M	Boundary router global = gate-bin: 29.02M, violation 0.3420, shortage 14.51M	Penalty is too low to buy reserve; gate-bin does not change the boundary policy.
5	Graph WaveNet/physical-bin: cost 60.62M, violation 0.2043, shortage 8.24M	Boundary router gate-bin 60.50M vs global 62.44M; violation 0.2199 -> 0.2029; shortage 8.60M -> 7.77M	Gate-bin begins to reduce boundary shortage and violations.
10	Graph WaveNet/physical-bin: cost 84.31M, violation 0.0931, shortage 3.40M	Boundary router gate-bin 84.58M vs global 88.13M; violation 0.1038 -> 0.0900; shortage 3.73M -> 3.19M	Main operating point: boundary gate-bin is competitive and lowers shortage relative to same-model global reserve.
20	Graph WaveNet/physical-bin: cost 109.41M, violation 0.0411, shortage 1.23M	Boundary router gate-bin 110.90M vs global 112.30M; violation roughly unchanged 0.0374 -> 0.0375; shortage 1.06M -> 1.02M	Benefit narrows; the policy mainly shifts reserve allocation inside the boundary band.
50	Graph WaveNet/physical-bin: cost 135.78M, violation 0.0234, shortage 0.51M	Boundary router gate-bin 142.73M vs global 136.80M; violation 0.0125 -> 0.0160; shortage 0.23M -> 0.37M	At very high penalty, the same-model global policy is safer and cheaper.

Model	n	Overall RMSE	Switch RMSE	NMI	ARI
Anchor-only router	5	234.24 $\pm$ 6.72	236.91 $\pm$ 7.16	0.8661 $\pm$ 0.1546	0.8869 $\pm$ 0.1683
MoE + L_bal + L_align + L_force	5	233.87 $\pm$ 2.84	234.87 $\pm$ 2.33	0.8340 $\pm$ 0.0958	0.8829 $\pm$ 0.0875
Context-supervised router	5	229.47 $\pm$ 6.95	230.00 $\pm$ 6.72	0.7479 $\pm$ 0.0461	0.8084 $\pm$ 0.0717

Table 5. Reviewer-facing spatial holdout summary.

MoE and the dense encoder is small and uncertain, about 0.009 (-0.061 to 0.085), and the expanded ERA5 baseline panel shows that even the best learned baseline still trails persistence. The efficiency rows add one more cost: the boundary-forced router is slower than the full physics-aligned MoE. The robustness evidence therefore supports the routing-semantics claim and keeps the accuracy claim bounded.

The strict replay, intervention, placebo, and time-forward tables below are the current robustness evidence for the corrected-router claim.

We also add negative-control and intervention audits for the strict-mask WTB gate. The five-seed strict-mask set compares the actual operating labels with labels that should not preserve the MPPT-to-pitch boundary: an 18-step temporal shift, a one-day temporal shift, node permutation, within-time node shuffling, and global label shuffling. The boundary-forced router keeps high agreement with the actual labels (NMI/ARI 0.8716 $\pm$ 0.0374 / 0.9166 $\pm$ 0.0332 in the placebo summary). Actual labels exceed global shuffling by delta NMI/ARI 0.8716 / 0.9166, one-day temporal shift by 0.8324 / 0.8656, and node permutation by 0.3289 / 0.2825; all bootstrap intervals are positive. The per-seed placebo table shows that this separation is not carried by one lucky run: all five seeds retain large actual-vs-global and actual-vs-one-day-shift gaps. Spatial placebos remain partially aligned because turbines share fleet-level operating distributions, but the actual labels are still clearly separated.

The mechanism intervention gives the stronger test. Reloaded strict-mask checkpoints match their saved metrics for all five runs before intervention. Zeroing the boundary anchor raises overall RMSE by 2.6792 (95% bootstrap interval 0.7347 to 4.5598), raises switch-window RMSE by 2.9470 (0.7985 to 5.1213), and drops NMI/ARI by 0.7017 / 0.8593. The per-seed intervention table sharpens the interpretation. Boundary-anchor removal collapses NMI/ARI in every seed, but the RMSE penalty is heterogeneous: seeds 202 and 205 show near-zero or slightly negative RMSE deltas while still losing most of their routing agreement. The mechanism claim is therefore about gate responsibility, not an assertion that removing the anchor always worsens headline error. Zeroing the wake score is a near-null control: overall RMSE changes by 0.0031 (-0.0016 to 0.0110), and NMI/ARI drop by about 0.0010 / 0.0010. This moves the WTB claim beyond correlation or visualization: the gate depends on the intended boundary anchor, not on any arbitrary physical channel.

Model	n	Overall RMSE	Switch RMSE	NMI	ARI
MoE + L_bal + L_align + L_force	5	235.49 $\pm$ 7.19	239.93 $\pm$ 7.27	0.8352 $\pm$ 0.0669	0.8870 $\pm$ 0.0669

Table 6. Reviewer-facing future-period holdout summary.

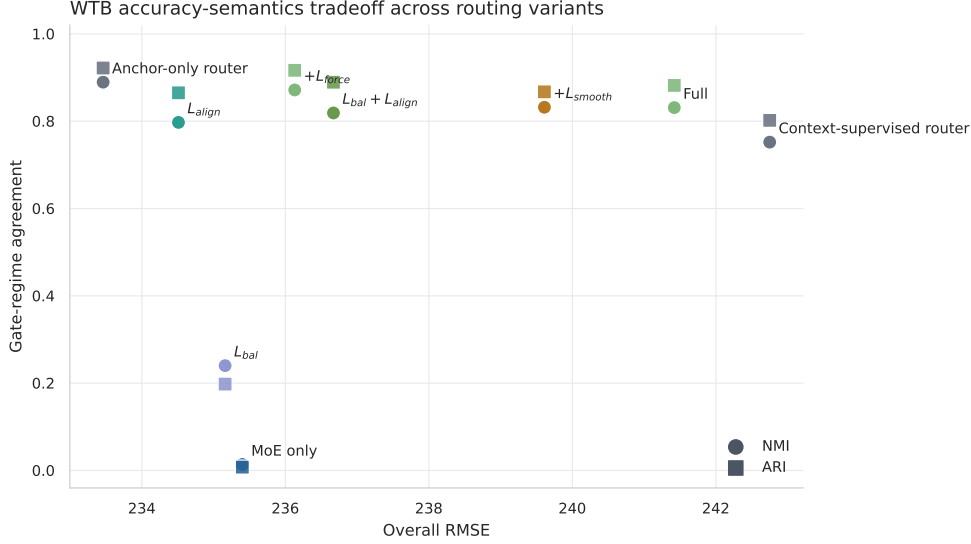


Figure 6: WTB accuracy-semantic tradeoff across routing variants. Circles show NMI and squares show ARI against overall RMSE. The frontier clarifies that stronger routing correction improves gate-regime agreement but can move the model away from the lowest forecast error.

The same evidence stack is traceable to source files in `artifacts/strict_wtb_evidence_sources_20260609`, which contains per-seed CSV files, generated LaTeX tables, and a SHA256 manifest for the input and output artifacts.

5.8 Time-forward stress test

A post-hoc time-forward test-slice audit asks whether the strict-mask evidence survives later test windows. The answer is split. Routing semantics remain stable: in the late test block, the corrected router keeps NMI/ARI 0.8661 $\pm$ 0.0393 / 0.9079 $\pm$ 0.0364, and late-minus-early NMI is +0.0068 with a 95% bootstrap interval from 0.0021 to 0.0127. Forecasting accuracy, however, worsens in the late block. Late-minus-early overall RMSE is +50.3454 (46.8137 to 54.1758), and switch-window RMSE is +63.3549 (60.7291 to 65.3319). The added distribution-shift diagnostics explain this as a boundary condition: the late block has higher mean target power (+52.70), a large regime-share shift (total variation 0.2983, Jensen-Shannon divergence 0.0539), lower mask-valid coverage (-0.1446), and a lower mean wind-speed anchor (-0.3028) but much larger pitch action (+13.9845 in mean Pab). The gate still tracks the physical partition, but the value map faces a different operating distribution. This audit therefore supports a routing-semantics claim while weakening any time-robust forecasting claim. It is a stress test on saved test predictions, not a replacement for an external dataset or a pre-specified future-period holdout.

5.9 Sensitivity checks

Table 8 tests whether WTB routing recovery depends on a single hand-picked boundary or one large loss weight. Across three-seed sweeps, changing λ_{align} from 2500 to 7500 keeps NMI in the 0.8255–0.8546 range, and changing λ_{force} from 5000 to 15000 keeps NMI in the 0.8091–0.8406 range. The new label-validity audit then re-labels the saved strict-cache gate outputs over the local grid rated-wind {10.0, 10.5, 11.0} and pitch-threshold {1.5, 2.0, 2.5} without retraining. The worst grid point still has N-

Table 7. Replay-safe strict-mask WTB mechanism interventions.

Intervention	Delta RMSE	Delta switch RMSE	Drop NMI	Drop ARI
anchor_boundary_zero	2.6792 [0.7347, 4.5598]	2.9470 [0.7985, 5.1213]	0.7017 [0.6451, 0.7911]	0.8593 [0.8208, 0.9075]
anchor_wake_zero	0.0031 [-0.0016, 0.0110]	0.0032 [-0.0011, 0.0106]	0.0010 [0.0001, 0.0021]	0.0010 [0.0002, 0.0023]
anchor_all_zero	7.2037 [2.1205, 12.2869]	7.7592 [2.2208, 13.2587]	0.7191 [0.6579, 0.8230]	0.8638 [0.8244, 0.9179]

Table 8. Per-seed strict-mask WTB mechanism intervention effects.

Seed	Intervention	Delta RMSE	Delta switch RMSE	Drop NMI	Drop ARI
201	anchor_boundary_zero	3.6255	3.9414	0.6710	0.8560
202	anchor_boundary_zero	-0.0719	-0.0840	0.8745	0.9466
203	anchor_boundary_zero	4.6187	5.2961	0.6312	0.8020
204	anchor_boundary_zero	5.1827	5.5364	0.6722	0.8418
205	anchor_boundary_zero	0.0408	0.0450	0.6597	0.8499
201	anchor_wake_zero	-0.0013	-0.0017	0.0005	0.0007
202	anchor_wake_zero	0.0006	0.0007	-0.0002	-0.0001
203	anchor_wake_zero	0.0187	0.0180	0.0011	0.0012
204	anchor_wake_zero	-0.0027	-0.0012	0.0033	0.0033
205	anchor_wake_zero	0.0001	0.0001	0.0002	0.0001

MI/ARI 0.8655 +/- 0.0429 / 0.9146 +/- 0.0377, and the minimum shared-valid support is 0.9909. These rows are threshold/label validity checks and mask controls, not the semantic negative control. The semantic negative-control claim is reserved for `boundary_negative_controls_wtb` and the routing placebo audit, where deliberately incorrect or shifted labels are checked against the same intervention evidence. The main WTB result keeps the pre-specified operating-boundary rule.

5.10 Claim boundary

The external Kelmarsh/Penmanshiel result is a failed boundary-condition test, not a new-site success. The source guard completes 80/80 runs, both leave-one-farm-out directions, and the chronological sanity checks, but the default WTB boundary gives mean routing NMI 0.4877, below the pre-specified 0.50 external-site routing criterion; the rescue guard therefore forbids portability wording and records the safe fallback as within-WTB boundary routing evidence with external boundary-condition diagnostics. We add local recalibration only to explain the failure mode. For each routing run, the validation gate output selects a farm-specific rated-wind/pitch-threshold pair from the pre-specified grid, and that selected boundary is evaluated once on the test gate output. This does not retrain the forecaster and does not rescue an out-of-site mechanism claim. Overall default test NMI rises only from 0.1324 to 0.1491 after recalibration, a recovery of +0.0167. Kelmarsh chronological runs recover more agreement, but Penmanshiel chronological and the leave-one-farm-out settings remain weak, so the failure is not merely a shifted threshold.

The small calibration-window adaptation makes that boundary sharper. Using only each run’s validation gate outputs, the diagnostic selects a local rated-wind/pitch pair and a majority-vote gate-to-regime map, then freezes both before evaluating held-out test gates. The guard completes 40/40 routing runs, but site-specific adaptation wording remains forbidden. Chronological adapted balanced accuracy averages 0.4787, below the 0.50 diagnostic threshold. Penmanshiel chronological improves from 0.4000 to about 0.5000 balanced accuracy, but Kelmarsh chronological declines and the leave-one-farm-out Penmanshiel-to-Kelmarsh setting collapses. The result is therefore a completed negative adaptation diagnostic: local boundary estimation is necessary, but this small window is not sufficient to restore robust usable routing at a new site.

The failure conditions are operationally informative. Kelmarsh has 6 Senvion MM92 turbines, where-

Table 9. Strict-mask WTB routing placebo separations.

Control	Delta NMI	Delta ARI
temporal_shift_18	0.5875 [0.5551, 0.6210]	0.5759 [0.5466, 0.6043]
temporal_shift_144	0.8324 [0.8020, 0.8692]	0.8656 [0.8345, 0.8954]
node_permutation	0.3289 [0.3016, 0.3659]	0.2825 [0.2581, 0.3100]
within_time_shuffle	0.3337 [0.3040, 0.3702]	0.2872 [0.2605, 0.3153]
global_shuffle	0.8716 [0.8411, 0.9086]	0.9166 [0.8879, 0.9422]

Table 10. Per-seed strict-mask WTB routing placebo separations.

Seed	Control	Actual NMI	Control NMI	Delta NMI	Delta ARI
201	global_shuffle	0.8685	0.0000	0.8685	0.9242
202	global_shuffle	0.9377	0.0000	0.9377	0.9691
203	global_shuffle	0.8221	0.0000	0.8221	0.8659
204	global_shuffle	0.8591	0.0000	0.8591	0.9059
205	global_shuffle	0.8708	0.0000	0.8708	0.9177
201	node_permutation	0.8685	0.5431	0.3255	0.2890
202	node_permutation	0.9377	0.5408	0.3968	0.3336
203	node_permutation	0.8221	0.5380	0.2841	0.2411
204	node_permutation	0.8591	0.5457	0.3134	0.2695
205	node_permutation	0.8708	0.5459	0.3249	0.2796
201	temporal_shift_144	0.8685	0.0400	0.8285	0.8732
202	temporal_shift_144	0.9377	0.0387	0.8990	0.9184
203	temporal_shift_144	0.8221	0.0383	0.7838	0.8146
204	temporal_shift_144	0.8591	0.0385	0.8205	0.8556
205	temporal_shift_144	0.8708	0.0405	0.8303	0.8663

as Penmanshiel has 14 retained Senvion MM82 turbines. Pitch-feature coverage and boundary support differ sharply, especially in leave-one-farm-out caches, and the regime shares move from WTB’s 39.47% idle, 52.00% MPPT, and 8.53% pitch-control partition to Kelmarsh’s MPPT-heavy chronological partition and Penmanshiel’s larger pitch-control share. These differences in sensor fields, pitch observability, turbine geometry, farm scale, power-curve distribution, and label balance are exactly the conditions under which the WTB operating-boundary rule must be re-estimated and tested before any site-level claim.

6 Discussion

The main result is a rule for using physics-aligned routing in wind-power operations: correction strength should scale with regime observability and with the operating decision that will consume the gate. ERA5 already exposes the regime marker through sensible heat flux and its temporal variation, and persistence is hard to beat on RMSE. In that setting, the corrected gate improves regime alignment over Unconstrained MoE without creating a better headline forecasting map. WTB is different. Wake interaction and blade-pitch control blur the MPPT-to-pitch boundary, so the router needs stronger repair to recover a useful partition. The WTB-ERA5 contrast is the paper’s main scientific insight: the same routing prior has different value depending on how much of the physical boundary is already visible in the observations.

Table 11. Strict-mask WTB post-hoc time-forward test-slice audit.

Block	Anchor range	Overall RMSE	Switch RMSE	NMI	ARI
1 early	30275-31520	217.4623 +/- 9.6060	209.3283 +/- 8.7584	0.8593 +/- 0.0420	0.9182 +/- 0.0325
2 middle_2	31521-32765	243.6546 +/- 7.4467	250.9040 +/- 9.6873	0.8693 +/- 0.0374	0.9117 +/- 0.0365
3 middle_3	32766-34010	199.7237 +/- 7.6958	200.7185 +/- 7.9983	0.8500 +/- 0.0445	0.9207 +/- 0.0318
4 late	34011-35255	267.8077 +/- 10.4767	272.6831 +/- 9.7232	0.8661 +/- 0.0393	0.9079 +/- 0.0364

Table 12. Strict-mask WTB late-period distribution-shift diagnostics.

Model	Delta RMSE	Delta abs ramp	Regime TV	Regime JSD	Delta NMI
MoE + L_bal + L_align + L_force	50.3454 [46.8137, 54.1758]	-1104.6378 [-1104.6378, -1104.6378]	0.2983	0.0539	0.0068 [0.0021, 0.0127]

The price is explicit. Graph WaveNet is the most stable black-box WTB forecaster, while LightGBM and XGBoost lag-feature baselines show that simple engineering predictors remain highly competitive. Inside the routed family, the full physics-aligned MoE is the forecasting compromise, while the boundary-forced router is the semantic-repair variant. Regime-anchor alignment buys a large share of the partition recovery with a modest RMSE change. Boundary forcing buys a cleaner gate at a clear forecasting cost. That is the paper’s practical claim: gate correction is a modeling choice with measurable error and reserve-policy tradeoffs.

For mechanism-aware forecasting studies, physics has to constrain the routing decision itself. Many physics-guided models act on outputs or latent states, which suits tasks where the goal is physical consistency after prediction. Here the decision is earlier: which local predictor takes responsibility near a mechanism change. The MPPT-to-pitch boundary is a physically meaningful stress point because it changes reserve shortage risk and ramp-risk interpretation even when fleet-level average error moves only modestly. Boundary-focused forcing encodes that stress point at the router. It sacrifices some RMSE to keep the gate from drifting into a semantically useless partition.

The method is appropriate when the operating boundary is part of the scientific question. Examples include turbine-control transitions, curtailment regimes, and geophysical stability states where the active mechanism matters, not only the scalar error. It is a poor choice for a pure leaderboard objective, for settings where persistence or a black-box graph model is already adequate, or for problems without a defensible regime anchor. In those cases, the extra routing losses add complexity without a clear mechanism-evidence return. The target use case is the middle ground: mechanism boundaries are important enough that interpretable responsibility can justify a measured forecasting penalty. This positioning also explains why the strongest claim is about controlled responsibility assignment rather than broad cross-dataset superiority.

7 Limitations

The WTB correction depends on threshold-based pseudo-labels derived from wind speed and mean pitch angle. Those thresholds are physically motivated and consistent with the implemented data pipeline, but they impose a hand-crafted boundary definition. The method is boundary-aware regularized routing.

The comparison is bounded. Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, persistence, physical power curve, XGBoost/LightGBM lag features, and a DLinear-style LTSF readout provide completed strict-cache WTB forecasting baselines, and ERA5 is checked against persistence plus Graph WaveNet, STGCN, PatchTST, and TCN. The strict-anchor-mask rerun covers the selected WTB corrected router, replay, intervention, placebo, time-forward audits, a synchronized baseline refresh, a spatial node-held-out stress test, a future-period holdout, and the transition-window reserve audit. Market-level and dispatch-level reserve optimization, unit commitment, stochastic reserve optimization, larger general time-series backbones, and broader graph-transformer variants are outside the present benchmark. External wind-farm diagnostics are complete but fail the external-site routing criterion: the Kel-

Table 8. WTB sensitivity to routing-loss weights and operating-boundary thresholds (mean $\pm$ std across three seeds).

Setting	Value	Overall RMSE	Switch RMSE	NMI	ARI
<i>Panel A. Routing-weight sensitivity</i>					
λ_{align}	2500	258.00 $\pm$ 12.40	261.37 $\pm$ 13.96	0.8402 $\pm$ 0.0505	0.9026 $\pm$ 0.0378
λ_{align}	5000	254.62 $\pm$ 4.56	258.47 $\pm$ 5.61	0.8255 $\pm$ 0.0058	0.8779 $\pm$ 0.0019
λ_{align}	7500	257.34 $\pm$ 2.90	262.83 $\pm$ 3.59	0.8546 $\pm$ 0.0427	0.9093 $\pm$ 0.0282
λ_{force}	5000	257.61 $\pm$ 2.43	261.66 $\pm$ 2.66	0.8406 $\pm$ 0.0237	0.8952 $\pm$ 0.0182
λ_{force}	10000	259.51 $\pm$ 5.60	262.66 $\pm$ 8.28	0.8152 $\pm$ 0.0456	0.8650 $\pm$ 0.0479
λ_{force}	15000	261.84 $\pm$ 2.63	266.45 $\pm$ 1.62	0.8091 $\pm$ 0.0102	0.8522 $\pm$ 0.0081
<i>Panel B. Operating-boundary threshold sensitivity</i>					
u_{rated}	10.0	256.45 $\pm$ 3.36	260.64 $\pm$ 4.99	0.8655 $\pm$ 0.0230	0.9151 $\pm$ 0.0223
u_{rated}	10.5	258.76 $\pm$ 5.51	261.93 $\pm$ 7.04	0.8375 $\pm$ 0.0466	0.8795 $\pm$ 0.0485
u_{rated}	11.0	251.71 $\pm$ 4.15	255.65 $\pm$ 4.96	0.8662 $\pm$ 0.0133	0.9179 $\pm$ 0.0081
p_{th}	1.5	255.31 $\pm$ 5.88	261.11 $\pm$ 5.56	0.8557 $\pm$ 0.0331	0.9031 $\pm$ 0.0313
p_{th}	2.0	253.41 $\pm$ 4.22	256.41 $\pm$ 7.23	0.8350 $\pm$ 0.0384	0.8786 $\pm$ 0.0386
p_{th}	2.5	258.28 $\pm$ 7.66	264.11 $\pm$ 7.77	0.8637 $\pm$ 0.0031	0.9144 $\pm$ 0.0040

Table. WTB threshold/label validity audit for saved strict-cache gate outputs.

Rated wind	Pitch threshold	Runs	NMI	ARI	Shared valid
10.0	1.5	5	0.8724 $\pm$ 0.0404	0.9165 $\pm$ 0.0368	0.9956
10.0	2.0	5	0.8721 $\pm$ 0.0407	0.9164 $\pm$ 0.0369	0.9991
10.0	2.5	5	0.8717 $\pm$ 0.0410	0.9163 $\pm$ 0.0370	0.9990
10.5	1.5	5	0.8723 $\pm$ 0.0418	0.9165 $\pm$ 0.0372	0.9960
10.5	2.0	5	0.8710 $\pm$ 0.0420	0.9161 $\pm$ 0.0374	1.0000
10.5	2.5	5	0.8696 $\pm$ 0.0422	0.9157 $\pm$ 0.0374	0.9998
11.0	1.5	5	0.8687 $\pm$ 0.0426	0.9156 $\pm$ 0.0375	0.9909
11.0	2.0	5	0.8672 $\pm$ 0.0429	0.9151 $\pm$ 0.0377	0.9949
11.0	2.5	5	0.8655 $\pm$ 0.0429	0.9146 $\pm$ 0.0377	0.9949

marsh/Penmanshiel guard reaches 80/80 completed runs in both leave-one-farm-out directions, while the default WTB boundary remains below the declared threshold. The added local recalibration table records how much agreement is recovered when rated-wind/pitch boundaries are re-estimated on validation gates and then fixed on test gates. The external taxonomy points to farm scale, turbine geometry, pitch observability, sensor-field coverage, power-curve differences, and regime-label imbalance as the main boundary conditions. The evidence therefore supports within-WTB spatial and temporal stress validation of routing semantics and operating-boundary reserve diagnostics, plus an external diagnostic of when those semantics need local boundary re-estimation.

The time-forward audit is not an external validation. It is a post-hoc split of saved WTB test predictions into contiguous anchor-time blocks. It shows that the strict-mask gate keeps its physical partition in the late block, but it also shows a large late-test RMSE increase. The failure case is therefore explicit: late-period distribution shift preserves routing agreement while degrading value prediction. This is an important boundary condition: the method currently supports routing semantics, not time-robust forecasting robustness.

The WTB ablation is scoped. The strict-cache 50-run ablation guard passes, but each cited row must report RMSE together with NMI/ARI. The ablation supports the observed accuracy-semantics tradeoff; it does not convert the routed model into the lowest-error predictor.

Table 9. Claim boundary for the current manuscript.

Claim	Status	Safe wording
WTB operating-boundary routing accountability	Supported	Boundary-forced routing recovers the MPPT/pitch partition under strict-cache WTB replay, intervention, placebo, threshold-validity, and within-WTB spatial/future stress tests.
Headline forecast-error dominance	Not supported	Graph WaveNet and lag-feature engineering baselines remain stronger WTB forecasters on RMSE; the routed model pays an explicit forecasting cost.
Boundary-window reserve relevance	Supported, conditional	Gate-aware reserve is a transition-window diagnostic that can reduce boundary-window shortage and violation in selected operating slices; it is not market-level, dispatch-level, or stochastic reserve optimization.
Failure-case transparency	Supported	Gate-correct but forecast-bad cases are explicitly reported to separate mechanism responsibility from value accuracy.
Late-period time robustness	Diagnostic only	Late WTB slices preserve routing semantics but show large RMSE degradation under distribution shift.
Kelmarsh/Penmanshiel external-site routing portability test	Not supported	The completed external runs fail the external-site routing criterion and are reported as boundary-condition diagnostics requiring local re-estimation.
Kelmarsh/Penmanshiel small-calibration adaptation	Diagnostic only, not restored	A validation-window boundary and gate-map adaptation is complete but does not restore robust usable routing overall; Penmanshiel chronological improves locally while Kelmarsh and leave-one-farm-out remain weak.

Statistical reliability is uneven across claim types. The completed strict-cache WTB baseline refresh provides five seeds each for Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST, and the selected boundary-forced router has five strict-cache seeds, five future-holdout seeds, and five seeds for each spatial-holdout routing comparator. ERA5 has five seeds for the in-family models and three seeds for the learned baselines. The engineering baselines and reserve-window audit are deterministic or sampled strict-cache diagnostics rather than five-seed neural families. Semantic negative controls are taken from `boundary_negative_controls_wtb`, and the mask/threshold sensitivity controls are reported as sensitivity checks only. The external Kelmarsh/Penmanshiel guard is complete at 80/80 but remains `within_wtb_only`; the recalibration and small-calibration-window adaptation diagnostics are therefore reported as boundary-condition analyses, not as new training results or external-site evidence. The strongest statistical claims are therefore the strict-mask replay-safe routing-recovery, mechanism-intervention, failure-case transparency, and within-WTB spatial/temporal stress claims.

The paper studies two observability settings. WTB and ERA5 are enough for the intended contrast between weakly and strongly expressed regime markers, with ERA5 serving as a positive/contrast control rather than as the main performance contribution. Applying the same interpretation to other industrial or geophysical systems will require re-estimating the operating boundary, checking sensor availability, and rerunning the routing gate before making a local mechanism claim.

8 Conclusion

This paper turns non-stationary wind-power forecasting into a routing problem around operating transitions. The completed strict-cache WTB comparison gives a blunt accuracy result: strong graph, transformer, and lag-feature baselines achieve lower mean RMSE than the boundary-forced routed model. The

strict-anchor-mask evidence then tests the selected router more severely: across five seeds it recovers the operating partition, passes checkpoint replay, fails under boundary-anchor removal, survives placebo controls, remains meaningful under spatial and future-period holdouts, and keeps routing semantics in late test windows. ERA5 carries the same lesson as an observability contrast: when the regime marker is visible and persistence is strong, physics-aligned routing improves gate-regime agreement more than headline forecasting accuracy.

The practical conclusion follows directly. When the regime marker is visible, routing regularization calibrates an already available partition. When the marker is partly hidden by control logic, physics guidance at the gate can repair a weakly identified partition and provide a transition-window reserve/ramp-risk signal, but it trades against forecast error and does not create a globally better reserve policy. Physical forecasting with multiple local response laws therefore needs supervision over responsibility as well as supervision over values. The completed external wind diagnostic shows that this responsibility claim must be re-established when turbine geometry, pitch observability, sensor fields, power curves, or label distributions change; a small validation-window adaptation did not restore robust held-out routing in Kelmarsh/Penmanshiel. The gate is part of the model’s scientific claim, and the right question is whether that claim is worth its measured error, reserve, robustness, and efficiency costs in the target operating setting.

9 Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT/Codex to support language editing, consistency checking, and submission-material drafting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

10 Code and data availability

The experiments use four public source families: the KDD Cup 2022 wind-farm SCADA benchmark, ERA5 reanalysis fields, and the Kelmarsh and Penmanshiel wind-farm SCADA records. Source datasets remain available from their original providers. The release package records provider links, checksums, and reconstruction scripts rather than redistributing any source files whose original terms require provider-side access. The Kelmarsh and Penmanshiel sources are tracked in the reproducibility manifest as CC-BY-4.0 source-data guards and protocols; their completed 80-run guard, recalibration table, and small-calibration-window adaptation table are cited only as boundary-condition evidence because new-site portability is not claimed. The final evidence export includes table, figure, source-data, guard, and trace manifests; each submission-facing table or figure is linked to source files, seed lists, run tables, raw prediction paths, checkpoint hashes, and SHA256 hashes where file size permits. Code, configuration files, scripts, derived caches when licensing permits, model-weight/checkpoint hashes, seed-level CSV summaries, final figure source data, and evidence manifests can be released with the manuscript. Raw third-party data that cannot be redistributed will be accompanied by download instructions and rebuilding scripts. The one-command reproduction package rebuilds the strict evidence package, the engineering baseline panel, the transition-window reserve audit, the reserve guard, the external recalibration and small-calibration diagnostics, the final evidence manifest, and the reproducibility manifest with relative paths. The study uses turbine and atmospheric measurements only and involves no human participants or human-subject data.

11 Appendix A. Training and implementation details

11.1 Training loop summary

The optimization procedure is identical across runs except for the routing penalties that are activated in each setting and the anchor labels that are available.

Algorithm A1 Physics-aligned MoE training

Input: history windows X , graph sequence A , targets Y , primary regime anchors R , auxiliary wake labels W (WTB only), model parameters θ

for each minibatch do

 Build node-level graph context from the precomputed graph tensors

 Encode X with the directed-diffusion GRU to obtain node context H

 Compute gate logits Z and soft routing probabilities G

 Compute expert forecasts and aggregate them with G

 Evaluate L_{pred}

 If MoE mode is active:

 add L_{bal} on top-K routing statistics

 add L_{align} on valid primary regime anchors

 if WTB: add L_{force} on MPPT/pitch samples

 if WTB: add L_{aux} on wake labels

 add L_{smooth} on the final graph snapshot

 Update θ with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

11.2 Shared constants

11.3 Dataset-specific thresholds and routing weights

Bai, L., Yao, L., Li, C., Wang, X., Wang, C., 2020. Adaptive graph convolutional recurrent network for traffic forecasting, in: *Advances in Neural Information Processing Systems*. pp. 17804–17815.

Daenens, S., Verstraeten, T., Daems, P.-J., Nowé, A., Helsen, J., 2025. Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data. *Wind Energy Science* 10, 1137–1152. <https://doi.org/10.5194/wes-10-1137-2025>

Fedus, W., Zoph, B., Shazeer, N., 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research* 23, 1–39.

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Table. Why external wind-farm portability failed and how the result is used.

Condition	Observed evidence	Why external-site routing failed	Safe use
Guard outcome	External guard completes 80/80 runs and both leave-one-farm-out directions, but mean routing NMI is 0.4877 below the pre-specified 0.50 external-site routing criterion; rescue summary forbids portability wording.	The evidence is complete enough to diagnose the boundary condition but does not clear the declared external-site routing threshold.	Report as complete external failure/condition diagnostic, not new-site success.
Local boundary recalibration	Overall default test NMI 0.1324 rises only to 0.1491 after validation-selected local rated-wind/pitch boundaries (+0.0167). Kelmarsh chronological recovers about +0.062 NMI; Penmanshiel chronological is essentially unchanged near zero.	Boundary re-estimation helps one chronological farm but does not recover leave-one-farm-out agreement; the failure is not only a threshold offset.	Use recalibration as an explanation of site-specific boundary conditions.
Small-calibration-window adaptation	The new adaptation guard completes 40/40 routing runs, but site-specific adaptation wording remains forbidden. Chronological adapted balanced accuracy averages 0.4787 below the 0.50 diagnostic threshold; Penmanshiel chronological improves from 0.4000 to about 0.5000, while Kelmarsh chronological declines and leave-one-farm-out Penmanshiel-to-Kelmarsh collapses.	A small validation window can select local boundaries and a gate-to-regime map but does not produce a stable held-out routing diagnostic across external sites or leave-one-farm-out directions.	Report as a completed negative adaptation diagnostic, not as recovery of external portability.
Sensor and pitch observability	Chronological pitch-feature coverage is 0.9446 for Kelmarsh and 0.6823 for Penmanshiel; leave-one support falls to 0.1066 for Kelmarsh-to-Penmanshiel and 0.0000 for Penmanshiel-to-Kelmarsh, with zero effective boundary cells in the latter.	The routing label can only transfer when the pitch and boundary observability used to construct it are present.	Require farm-specific sensor-field audits before applying WTB-derived operating labels.
Turbine geometry and farm scale	Kelmarsh has 6 Senvion MM92 turbines; Penmanshiel has 14 Senvion MM82 turbines. The rotor-diameter ratios between the two external sites are 0.891 and 1.122, while WTB contains 134 turbines under a compressed SCADA tensor.	A single WTB rated-wind/pitch rule cannot be assumed invariant across turbine geometry and fleet layout.	Treat rated wind, pitch threshold, and boundary-band support as local operating parameters.
Regime-share shift	WTB strict labels are 39.47% idle, 52.00% MPPT, 8.53% pitch. Kelmarsh chronological is 13.03% idle, 80.33% MPPT, 6.64% pitch; Penmanshiel chronological is 13.94% idle, 71.71% MPPT, 14.35% pitch.	The operating prior changes: one site is MPPT-heavy, another has a much larger pitch-control share, and leave-one support can collapse.	Use external runs to map the operating envelope that must be re-estimated.

Table A1. Shared architecture, graph, and training constants used in the reported experiments.

Item	WTB	ERA5	Role
History / horizon (H, P)	(36, 24)	(36, 24)	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top- K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius d_{max}	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

Table A2. Dataset-specific regime thresholds used to construct routing anchors.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s ⁻¹ , idle / active split
WTB	u_{rated}	10.5 m s ⁻¹ , MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of $ \text{sshf} $ on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split

Table A3. Active routing-loss weights in the reported corrected models.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence